NAVEEN RAJ DHANAPAL

Austin, TX | Phone: 301-300-9329 | E-mail: naveen.flash@gmail.com | LinkedIn: https://www.linkedin.com/in/naveenrd/

Staff Software Engineer with a strong background and hands-on experience in: *Application Development and Platform Engineering Leadership,* specializing in the architecture, design, and development of resilient, fault-tolerant, and scalable enterprise software and AI agentic applications.

- **AI & Platform Architecture:** Currently driving the control and data plane platform for the Vehicle Data Cloud to deliver remote vehicle configurations, bridging high-scale distributed systems with RAG architectures and LLM capabilities to orchestrate intelligent agentic workflows.
- **Federated API Strategy:** Pioneered the design and implementation of a Backend-for-Frontend (BFF) layer using GraphQL Managed Federation, leveraging Supergraph-Subgraph architecture to enable efficient data access for enterprise Web and Mobile applications.
- **Cloud-Native Engineering:** Strong passion and expertise in architecting resilient, multi-cloud environments (Azure, AWS, GCP) utilizing Java, Go, Python, and TypeScript, with a deep focus on containerized orchestration (Docker, Kubernetes) and Infrastructure as Code (Terraform).
- **Technical Leadership:** Spearheaded cross-functional teams to drive multi-platform initiatives, enforcing enterprise schema governance, system performance standards, & driving innovation in cloud-native technologies.
- **Mentorship & Culture:** Dedicated to mentoring junior software engineers, promoting best practices, fostering a culture of technical excellence, and serving as a featured speaker on modern engineering practices.

TECHNICAL EXPERTISE

Languages - Java, Python, TypeScript - Node.js, Go, Groovy, React
Databases - SQL (PostgreSQL, MySQL) & NoSQL (Cosmos DB, Firestore, Neo4j, DynamoDB, MongoDB, Redis)
Frameworks - Spring Framework, ApolloServer, Kafka, Flink, Akka, gRPC, Protobuf, WebSockets
Infrastructure & Observability - Docker, Kubernetes (AKS), Terraform, OpenTelemetry (OTel), Datadog, Splunk, Elasticsearch (ELK), GitHub Actions

PROFESSIONAL EXPERIENCE

Staff Software Engineer *(October 2025 – Present)* **General Motors – Austin, TX**

- Architecting and scaling the core Control and Data Plane, powering remote vehicle configuration and enabling secure, high-volume data ingestion and orchestration.
- Building custom Model Context Protocol (MCP) tools and autonomous, stateful agentic workflows that enable LLMs to interact directly with distributed enterprise data layers.
- Architected and deployed resilient, highly scalable backend microservices utilizing Spring Boot, Spring Modulith, and Spring AI, orchestrating complex containerized workloads on AZ – AKS.
- Engineered highly scalable event-driven architectures leveraging Azure Event Hubs and Service Bus, orchestrating massive-scale asynchronous messaging for vehicle telemetry and configuration data.
- Established comprehensive CI/CD and observability frameworks using Terraform, GitHub Actions, Datadog, and OpenTelemetry to ensure high availability and performance of critical cloud infrastructure.

Principal Software Engineer *(January* 2025 *– October 2025)* **Comcast Corporation – Austin, TX**
Lead Software Engineer *(September 2019 – December 2024)*

- Architected the federated GraphQL platform for the Xfinity App, establishing enterprise schema governance, performance standards, and long-term API evolution across the Account, Billing, and Payments domains.
- Developed and integrated a GraphQL Managed Federation Architecture using Apollo Server, Neo4j DB, enabling seamless collaboration between multiple microservices (aka. Subgraphs).

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- Spearheaded the design and implementation of distributed GraphQL services to power Payment Systems, enabling efficient and real-time data retrieval to handle distributed queries across multiple microservices.
- Leveraged event-driven architecture with Kafka and AWS SQS to enable asynchronous processing for high-volume payment, notification events, ensuring system reliability and scalability.
- Orchestrated the design and implementation of Spring Webflux microservices – Region of Interest that accepts camera motion events from Kafka via Apache Flink to calculate intersection with user-defined camera regions.
- Designed and implemented integration with AWS Rekognition and Sagemaker ML models to detect and tag objects from camera events, enhancing real-time event processing and improving object recognition accuracy.

Senior Software Developer *(September 2014 – September 2019)* **Rackspace Hosting Inc – San Antonio, TX**

- Led the design and development of PCI/SOX-compliant Payment Services and comprehensive Billing applications, integrating directly with the Chase Orbital Gateway and Oracle BRM.
- Founding engineer built the core Billing Services, integrating with Oracle BRM for billing operations, including invoicing, payments, adjustments, refunds, reversals, write-offs, and discounts.
- Implemented the Repose API Gateway across the billing layer to enforce strict RBAC.
- Developed Payment Service Web App, an MVC-based web form integrated into the Rackspace customer portal, enabling users to securely add and manage payment - Credit Cards and Electronic Checks.
- Designed and developed Offer Service for pricing, providing an enterprise product catalog with detailed pricing information for Rackspace's product offerings.

Software Developer II *(March 2012 – August 2014)* **Garmin International Inc – Olathe, KS**

- Agile scrum developer involved in the design, development and maintenance of various Garmin's Intranet/Extranet web applications, and REST based Web Services.
- Performed physical and logical data modeling, performance tuning – MySQL, SQLServer.

Software Developer Intern *(May 2011 – August 2011)* **Rackspace Hosting Inc – San Antonio, TX**

- Performed multi-tiered application development using J2EE standards (HTML, JavaScript, JSP, Java Servlets, JavaBeans, JDBC), Xerces parser, Structs and Spring Framework for Restful Web Services.
- Developed a client system using Jersey client, Spring MVC and the WADL parser using SAX/DOM.

EDUCATION

MASTER OF SCIENCE – COMPUTER SCIENCE Oklahoma State University
BACHELOR OF ENGINEERING – COMPUTER SCIENCE Anna University, India
POSTGRAD – AI & ML BUSINESS APPLICATIONS University of Texas at Austin, McCombs School of Business

CERTIFICATIONS

Model Context Protocol (MCP) Certification – Anthropic | Apollo Graph Developer Associate - Apollo GraphQL | AWS Certified Solution Architect Associate | Microsoft Certified: Azure Fundamentals | CKAD - Certified Kubernetes Application Developer | Certified Spring Professional v5.0 - Credential ID: 369542

VOLUNTEER

Technical Mentor, CodePath

- Volunteered to support college students through CodePath's Summer Technical Interview Prep course.
- Helped students to solve Data Structure and Algorithm problems in virtual whiteboard & coding interviews.